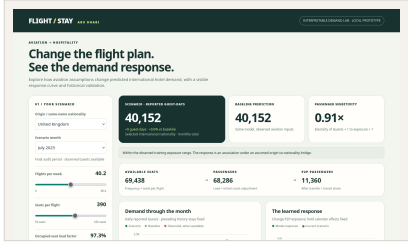
PROPOSAL / WORKING PROTOTYPE

Flight / Stay

A flight-to-hotel scenario lab for Abu Dhabi.

EXECUTIVE SUMMARY

We built an offline simulator for existing origin markets. Change flights, seats, load factor or connecting share; inspect the demand response.
On an untouched June–July 2025 audit, nationality-level error fell from **35.0%** for a calendar-only baseline to **27.7%**. The slider response still needs causal validation.



Live prototype. Full view on slide 8.

Prepared by [Team name] ů [University / universities] ů October 2026

Who we are and what we built

STUDENT TEAM

[Name 1] Data preparation and checks

[Name 2] Modeling and validation

[Name 3] Interface and demonstration

Replace these placeholders with the actual team, roles and university

before submission.

CURRENT STATUS

- Data audit and model comparison notebooks run end to end.
- Five interpretable candidates compared on a forward time split.
- A working HTML simulator runs locally without a server.
- Python predictions and browser slider outputs are checked for agreement.

Our aim is a useful first tool that a planner can question, not an unexplained forecast.

WHY THIS MATTERS

If flights change, what happens to hotel demand?

Flights and hotels are planned from different datasets. A seat may carry a transfer passenger, a returning resident or a hotel guest. Direct seat-to-room conversion would mislead.

QUESTIONS TO SCREEN

- What if weekly frequency rises or falls?
- What if an airline uses a larger aircraft?
- Which months and markets respond most in the historical data?

The challenge asks for a working, transparent, historically tested scenario tool. Source: ATP 2026 DCT Abu Dhabi challenge statement.

The data need an honest bridge

WHAT THE FILES CONTAIN

- Daily flight seats, passengers and connections; daily hotel *Guests* by nationality.
- 45 hotel nationalities; 33 have a same-name flight origin.
- Hotel *Guests* labels end in July 2025.

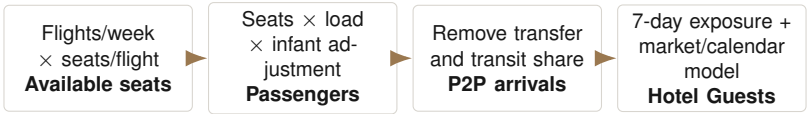
OUR STATED ASSUMPTIONS

- Same-name origin-to-nationality matching is a provisional link.
- The other 12 nationalities use a calendar forecast, with aviation controls disabled.
- Monthly *Guests* sums are reported guest-days. Capture, length of stay and occupancy are not separately identified.

Missing targets are excluded. We start in 2023 because 2022 flight observations have mixed time grain.

How the simulator works

VISIBLE ARITHMETIC, THEN A FITTED RESPONSE



INPUTS

Market, month, weekly frequency, seats per flight, load factor and transfer/transit share.

OUTPUTS

Predicted reported guest-days, change from observed-flight baseline, daily profile, learned response and extrapolation warning.

Visitor-to-hotel conversion is absorbed in the fitted response; its individual rate is not identified.

We chose the simpler model on evidence

FIVE CANDIDATES

1. Calendar-only baseline
2. Proportional passenger chain
3. Monotonic regression **(selected)**
4. Monotonic GAM, 7-day exposure
5. Monotonic GAM, stay-kernel exposure

The GAMs were slightly worse on the selection period. Extra curve complexity has not earned its place yet.

READABLE RESPONSE

$$\log(1+\widehat{G}) = \alpha_m + \beta_m \log(1+P_7) + \text{month} + \text{weekday} + \text{trend}$$
$$\beta_m \geq 0$$

A separate coefficient for each linked market keeps the passenger response non-decreasing. The seven-day passenger mean lets effects build through the month.

Selection uses Jan–Apr 2025 only. We favor the simpler eligible model within 1 percentage point of the best WMAPE.

What the back-test actually says

LOWER WMAPE IS BETTER

	Calendar	Flight model
Jan–Apr 2025 selection	31.9%	**25.0%**
Jun–Jul 2025 audit	35.0%	**27.7%**

WMAPE $= \sum |\text{prediction} - \text{observed}| / \sum \text{observed}$ across nationality-days. Models fit Jan 2023–Dec 2024 after a 27-day warm-up. May 2025 is reserved for error calibration.

OUR READING

7.2 points

Less error on the untouched audit.

27.7%

Too much error for precise market claims.

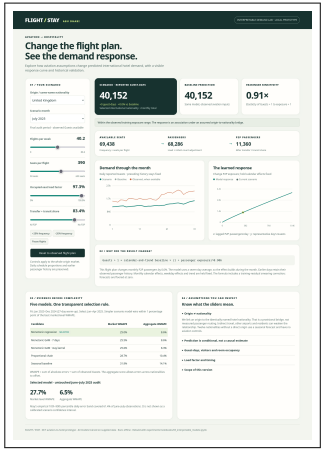
Aggregate daily WMAPE is 6.5% because market errors can cancel. Validation uses observed flights, not randomized changes.

A working prototype, not just a diagram

WHAT A PLANNER CAN DO NOW

1. Pick an existing origin market and month.
2. Move a slider for frequency, aircraft seats, load factor or connecting share.
3. Compare scenario and observed-flight baseline; inspect the passenger chain and demand curve.
4. See a warning if the scenario leaves the training range.

The prototype is a single offline HTML file. Model export and browser calculations are checked against Python.



Current HTML prototype, UK / July 2025.

Where this could be useful

INTENDED IMPACT

- Make flight and hotel planning assumptions visible in one place.
- Compare existing-market scenarios quickly before deeper analysis.
- Flag high-sensitivity months and markets for human review.

REAL-LIFE REQUIREMENTS

- Confirm the hotel's *Guests* definition and obtain room inventory for occupancy.
- Improve the departure-origin to nationality bridge with approved aggregated data.
- Refresh source data and monitor errors by market and season.
- Gate unsupported new routes until a defensible prior is available.

The current slider output is an association under stated assumptions; it is not a verified causal effect of adding flights.

If shortlisted: test the hard part

DEMONSTRATION AND VALIDATION PLAN

1. Demo a baseline, a 20% frequency increase and a flight cut in the same market/month.
2. Ask DCT to confirm target definitions and the origin-to-nationality mapping.
3. Predefine new time periods and actual route-change cases; test model error and the direction of change.
4. Report failures by market, season and extrapolation level; revise only with a fresh untouched period.

WHAT WE NEED TO IMPROVE

Aggregated origin–nationality flows, confirmed hotel guest-night definitions, event dates and room inventory. These would let us test more of the requested seats-to-visitors-to-nights chain.

Prototype and notebooks are ready for a live walkthrough. Team details and final video link should be inserted before submission.

Sources: ATP 2026 DCT Abu Dhabi challenge statement (challengeon.atrc.ae); supplied DCT aviation and hotel files; our executed domain analysis and model-comparison notebooks. Results reproduced 11 October 2026.